

Chapter 1

Project Proposal

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1.1 Introduction and Description

Multiple sclerosis is a polygenic, complex autoimmune disorder where the immune system attacks myelin, the protective cover of nerve cells (axons) in the central nervous system (CNS). Although damage can be mitigated to an extent by the re-routing of messages using undamaged nerve cells (plasticity), the damage eventually becomes too large to compensate resulting in progressive neurodegeneration.

Gene expression is the process outlined by the central dogma of molecular biology: DNA is transcribed into RNA, and RNA is translated into functional products. The measure of how much a gene is expressed at a given time is referred to as its gene expression level, and the analysis of differential expression levels when comparing healthy people and diagnosed patients can aid in the development of disease signatures based on changes in gene activity. Advanced, high-throughput technologies for studying gene expression, such as single-cell RNA sequencing (scRNA-seq) and single-nucleus RNA sequencing (snRNA-seq), allow gene expression to be measured at the level of individual cells or nuclei, enabling the heterogeneity of cell populations to

be explored. Studies have utilised snRNA-seq data in clustering distinct MS sub-groups [2] providing a basis for use in future individualised treatment and prediction.

Given that the complexity of biological networks in higher organisms primarily arises from dynamic gene interactions as opposed to the size of the genome (C-value paradox), analysing gene dynamics offers a more informative framework for understanding biological systems. Gene-gene co-expression networks (GGCNs) are typically presented as undirected graphs where nodes are genes and the presence of edges depends on co-expression values calculated from gene expression data.[5] These networks can be used to identify the functional status of genes from a systematic perspective, which can in turn inform the discovery of gene-disease associations through guilt-by-association analyses.

Graph neural networks (GNNs) have commonly been used for tasks like graph classification, node classification and link prediction.[4] Given a graph $G(V, E)$ as input in the form of a node feature matrix ($H : \mathbb{R}^{N \times F}$), GNNs produce new node embeddings that capture the node features, and the learnt embeddings can be used for graph classification. Graph Attention Networks (GATs) introduce a masked attention mechanism which take the neighbourhood of a node into account in computing the updated features for a node[6] - previous works have successfully applied these networks to disease prediction[3], providing precedent for its use in the project.

The project is composed of two main goals:

- Using the hdWGCNA framework to construct multi-layer co-expression networks from snRNA-seq gene-expression data
- Constructing a Multiplex Graph Attention Network (mGAT) which takes the GGCNs as input and outputs a binary classification for MS with a structure-blind multilayer perceptron (MLP) model and a vanilla Graph Convolution Network (GCN) to serve as baseline comparisons[1].

1.2 Starting Point

I have some experience with machine learning beyond the Computer Science Tripos Part IA and Part IB, however this is primarily related to the baseline models for this project (MLP) and models unrelated to this project.

1.3 Success Criterion

1.4 Extensions and Further Work

As extension work, the aim is to implement a multi-omic approach to classification with the incorporation of scATAC-seq data with an intermediate fusion approach.

1.5 Work Plan

Michaelmas Term

- **Weeks 1-2 (6/10-19/10)**
Submit proposal drafts
- **Weeks 3-4 (20/10-2/11)**
- **Weeks 5-6 (3/11-16/11)**
- **Weeks 7-8 (17/11-30/11)**

1.6 Resource Declaration

I will be using my personal laptop, a 2024 Dell Inspiron 16 2-in-1, with 16GB memory and a 1TB SSD for software development. As a backup, I will use my secondary personal laptop with an external 1TB storage. I will continuously backup my code and dissertation with Git version control. I accept full responsibility for this machine and I have made contingency plans to protect myself against hardware and/or software failure.

In terms of software, I will use:

- hdWGCNA R package
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The project also requires the use of biological datasets for training data. These datasets have been taken from <https://zenodo.org/records> which are

publicly available to anyone.

Bibliography

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- [5] Sipko Van Dam et al. “Gene Co-Expression Analysis for Functional Classification and Gene–Disease Predictions”. In: *Briefings in Bioinformatics* (Jan. 10, 2017), bbw139. ISSN: 1467-5463, 1477-4054. DOI: 10.1093/bib/bbw139. URL: <https://academic.oup.com/bib/article-lookup/doi/10.1093/bib/bbw139> (visited on 09/14/2026).
- [6] Petar Veličković et al. *Graph Attention Networks*. Version 3. 2017. DOI: 10.48550/ARXIV.1710.10903. URL: <https://arxiv.org/abs/1710.10903> (visited on 09/11/2026). Pre-published.